

D1A

$$f(x) = x^2 + 1$$

$$g(x) = e^{-x}$$

$$g(f(x)) = e^{-(x^2+1)} = e^{-x^2-1}$$

D2A

$$f(x) = \begin{cases} x^2 & -1 \leq x \leq 0 \\ -\ln(x+1) & 0 < x < 1 \end{cases}$$

$$y = x^2$$

$$y' = 2x, \quad 2x \geq 0 \Rightarrow x \geq 0$$

$$g = -\ln(x+1)$$

$$g' = -\frac{1}{x+1}, \quad -\frac{1}{x+1}, \quad x+1 > 0 \Rightarrow x > -1$$

D3A

$$\lim_{n \rightarrow +\infty} \frac{x^{-n^2} + n \sqrt{n} - n^2}{(3-n)(4 \ln(n) - 2n + 1)} \sim \frac{-n^2}{12 \ln(n) - 6n + 3 - 4n \ln(n) + 2n^2 + 1} \sim \frac{-n^2}{2n^2} = -\frac{1}{2}$$

D7A

$$f(x) = \frac{x^2}{2} - e^{-3x}$$

$$f' = \frac{(2x)(2) - (x^2)(0)}{4} - (-e^{-3x})(-3) =$$

$$f''(x) = 1 + 3e^{-3x} \cdot (-3) = 1 - 9e^{-3x}$$

$$1 - 9e^{-3x} < 0 \Rightarrow -9e^{-3x} < -1 \Rightarrow 9e^{-3x} > 1 \Rightarrow e^{-3x} > \frac{1}{9} \Rightarrow -3x > \ln(1/9) \Rightarrow -3x > \ln(1) - \ln(9) \Rightarrow -3x > -\ln(9)$$

$$-3x > -\ln(9) \Rightarrow 3x < \ln(9) \Rightarrow x < \frac{\ln(9)}{3}$$

D 8A

$$\int_0^{\pi/2} x \cos x = x \sin x - \int \sin x = [x \sin x + \cos x]_0^{\pi/2} = \left(\frac{\pi}{2} \cdot 1 + 0\right) - (0 + 1) = \frac{\pi}{2} - 1$$

D 9A

$$\sum_{n=1}^{+\infty} \frac{n^2}{n!} \quad \lim_{n \rightarrow +\infty} \frac{n^2}{n!} \quad \text{propre à l'infini}$$

$$\frac{u_{n+1}}{u_n} = \frac{n^2+1}{(n+1)!} \cdot \frac{n!}{n^2} = \frac{n^2+1}{\cancel{n!} (n+1)} \cdot \frac{\cancel{n!}}{n^2} = \frac{n^2+1}{(n+1)n^2} = \frac{n^2+1}{n^3+1} \sim \frac{n^2}{n^3} = \frac{1}{n} = 0$$

la série converge

D 5A

$$f: [0, 2] \rightarrow \mathbb{R}, \quad f(x) = x^3 - 3x + 1$$

$$y = x^3 - 3x + 1 \quad f(1) = 1 - 3 + 1 = -3 + 2 = -1$$

$$y' = 3x^2 - 3 \quad f(2) = 2^3 - 3 \cdot 2 + 1 = 8 - 6 + 1 = 3 \quad \text{MAX}$$

$$3x^2 - 3 = 0$$

$$3x^2 = 3$$

$$x^2 = 1$$

$$x^2 = \pm 1$$

$$f(0) = 0 - 3 \cdot 0 + 1 = 1$$

D 6A

Ej 3A

$$y = \frac{x+2}{x(x+1)}$$

$$\frac{A}{x} + \frac{B}{(x+1)}$$

$$\int \frac{x+2}{x(x+1)}$$

$$\frac{A(x+1) + Bx}{x(x+1)}$$

$$\int \frac{2}{x} - \frac{1}{x+1}$$

$$Ax + A + Bx$$

$$2 \ln(x) - \ln(x+1) + C \quad \begin{cases} A+B=1 \\ A=2 \end{cases} \quad \begin{cases} 2+B=1 \\ A=2 \end{cases} \quad \begin{cases} B=1-2 \\ A=2 \end{cases} \quad \begin{cases} B=-1 \\ A=2 \end{cases}$$

$$\left[2 \ln(x) - \ln(x+1) \right]_2^4 = (2 \ln 4 - \ln 5) - (2 \ln 2 - \ln 3) = \cancel{2} \ln 2 - \ln 5 - \cancel{2} \ln 2 + \ln 3 = 2 \ln 2 - \ln 5 + \ln 3$$

Dato f, g con $g(x) \neq 0$ si dice che f è o-piccolo di g per $x \rightarrow +\infty$ se $\lim_{x \rightarrow +\infty} f(x)/g(x) = 0$

$$\lim_{x \rightarrow +\infty} \frac{2 \ln 2 - \ln 5 + \ln 3}{x} = 0 \quad \begin{matrix} \text{forochia} \\ \text{infiniti} \end{matrix}$$

2.1A

$$f(x) = \frac{x}{x^2+1} \quad \frac{x}{x^2+1} - \frac{1}{2} \leq 0 \Rightarrow \frac{2x - x^2 - 1}{2(x^2+1)} \leq 0 \Rightarrow \begin{cases} 2x - x^2 - 1 \leq 0 \\ 2(x^2+1) > 0 \end{cases} \begin{cases} -\frac{(x-1)^2}{4} \geq 0 \\ \forall x \end{cases} \Rightarrow \forall x$$

$$A(x) = \lg\left(\frac{x}{x^2+1}\right)$$

Dominio

$$\frac{x}{x^2+1} > 0 \quad \begin{cases} x > 0 \\ x^2+1 > 0 \end{cases} \begin{cases} x > 0 \\ A_x \end{cases} \Rightarrow \boxed{x > 0}$$

Lim

$$\lim_{x \rightarrow +\infty} \lg\left(\frac{x}{x^2+1}\right) = \lg(0) = -\infty \quad \text{asintoto obliquo ???}$$

$$\lim_{x \rightarrow +\infty} \frac{\lg\left(\frac{x}{x^2+1}\right)}{x} = 0 \quad \text{forcella inf} \quad \text{no asymptoto obliquo}$$

$$\lim_{x \rightarrow 0^+} \lg\left(\frac{x}{x^2+1}\right) = \lg\left(\frac{0^+}{0^++1}\right) = \lg(0) = -\infty \quad \text{asintoto verticale}$$

monotonic

$$y = \frac{x}{x^2+1}$$

$$y' = \frac{x^2+1 - (x)(2x)}{(x^2+1)^2} = \frac{x^2+1 - 2x^2}{(x^2+1)^2} = \frac{-x^2+1}{(x^2+1)^2}$$

$$f'(x) > 0 \Rightarrow -x^2+1 > 0 \Rightarrow -x^2 > -1 \Rightarrow x^2 < 1 \Rightarrow -1 < x < 1 \Rightarrow x < 1$$

$$f(x) = 0 \quad -x^2+1 = 0 \Rightarrow -x^2 = -1 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow x = 1$$

$$f(1) = \text{max}$$

$$f(0, +\infty) = (-\infty, \lg(1/2)]$$

E 2A

$$a_n = (n^2 + 3)^{1/n}$$

$$\lim_{n \rightarrow +\infty} (n^2 + 3)^{1/n} = \frac{\lg(n^2 + 3)}{n} = e^0 = 1$$

Dato 2 serie $\sum a_n, \sum b_n$ 2 serie a termini positivi

se $0 < a_n \leq b_n$ e b_n converge anche a_n converge

se $0 < a_n < b_n$ e a_n diverge allora anche b_n converge

$$\sum \frac{\lg(n^2 + 3)}{n} \quad b_n = \frac{1}{n}$$
$$0 < 1/n < \frac{\lg(n^2 + 3)}{n}$$

$1/n$ diverge quindi anche $\frac{\lg(n^2 + 3)}{n}$ diverge